

Coherence-Dependent Backreaction in Semiclassical and Analog Gravity: Controlled Limits and an Analog-Gravity Test

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We study an effective-field-theory extension of semiclassical gravity in which a coarse-grained coherence functional $\mathcal{C}[g, \rho; \mu]$, built from relative entropy, generalized entropy, and local geometric counterterms, generates an informational stress tensor $\Theta_{\mu\nu}$. The construction is presented as a controlled near-equilibrium framework rather than a UV-complete theory. In a 1+1D conformal toy model and in weak-field/Rindler limits, we obtain leading-order expressions that reduce to standard semiclassical behaviour when $\rho \approx \sigma[g]$. We then formulate an acoustic implementation for a Bose-Einstein-condensate sonic horizon. Within this implementation, coherent phonon injection yields a predicted density modulation $\delta\rho/\rho_0 \sim 10^{-6}$ near the horizon, whereas a thermal control state matched in injected energy gives no leading-order informational contribution. The density-response relation used for this observable is benchmarked to the standard hydrodynamic compressibility limit of Gross-Pitaevskii/Bogoliubov theory, while the informational source term itself remains phenomenological. A simple numerical model implementing the same scaling ansatz gives $A = \max_{|r-r_H| \leq L_{\text{coh}}} |\delta\rho/\rho_0| \approx 1.4 \times 10^{-6}$ for representative parameters, and one-at-a-time $\pm 25\%$ scans about a Gross-Pitaevskii-compatible baseline keep the coherent signal above 10^{-7} . A null result below 10^{-7} in this protocol would falsify the acoustic implementation studied here, not the broader EFT proposal by itself. A positive analog result would test the proposed state-geometry feedback structure, not by itself establish direct gravitational backreaction in astrophysical spacetime.

I. INTRODUCTION

Can analog-gravity experiments test whether geometry responds only to mean stress-energy or also to coarse-grained state mismatch? We study that question within an effective semiclassical framework and use a Bose-Einstein-condensate (BEC) sonic horizon as the first concrete test bed.

Our main phenomenological result is an acoustic implementation in which coherent phonon injection near the sonic horizon yields a density modulation $\delta\rho/\rho_0 \sim 10^{-6}$, while a thermal control state matched in injected energy gives no leading-order informational contribution. In the present paper this is the primary falsifiable prediction: a null result below 10^{-7} would falsify the specific acoustic implementation studied here.

A. The Physical Question

Semiclassical gravity uses the renormalized expectation value $\langle T_{\mu\nu} \rangle$ as a source in Einstein's equations [1]. That framework is tightly constrained and extremely successful, but it leaves open a narrower EFT question: can coarse-grained state mismatch relative to a geometry-adapted reference state generate additional backreaction terms in controlled near-equilibrium regimes?

Several lines of work motivate that question. Thermodynamic derivations of Einstein's equations [2] and

entanglement-based formulations in quantum field theory and holography implicate entropy and information geometry in the structure of spacetime. Penrose [3] and Diósi [4] proposed that gravity may couple to quantum superposition in ways not exhausted by mean stress-energy alone. Relativistic quantum-information studies likewise show that acceleration and curved backgrounds modify coherence properties of quantum states.

B. Our Approach and Main Results

We construct a *coherence functional* $\mathcal{C}[g, \rho; \mu]$ that measures the informational mismatch between a quantum state ρ and a geometry-adapted reference state $\sigma[g]$. Variation with respect to the metric defines an *informational stress tensor* $\Theta_{\mu\nu}$ that augments the semiclassical Einstein equations:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G(\langle T_{\mu\nu} \rangle_\rho + \Theta_{\mu\nu}[g, \rho; \mu]). \quad (1)$$

When $\rho \approx \sigma[g]$, we have $\Theta_{\mu\nu} \approx 0$ and standard semiclassical gravity is recovered. When ρ departs from the chosen reference state, additional state-dependent corrections can appear.

The framework, which we call *Coherism*, yields the following concrete results:

- Controlled limits:** In a 1+1D conformal toy model and in weak-field/Rindler limits, we obtain leading-order expressions for $\Theta_{\mu\nu}$ and recover the standard semiclassical limit when $\rho \approx \sigma[g]$.
- Geometry-dependent open-system structure:** The main text uses a phenomenological coarse-grained Lindblad ansatz, and Appendix F shows

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how geometry-dependent GKLS evolution arises in a standard Unruh-DeWitt detector model.

3. **Primary falsification path:** In the acoustic implementation, coherent phonon injection near a sonic horizon yields $\delta\rho/\rho_0 \sim 10^{-6}$, while a matched thermal control gives no leading-order informational contribution.
4. **Null test:** A result below 10^{-7} for the primary density-modulation observable would falsify the acoustic implementation studied here.

C. Framework Overview

The basic picture is sketched in Fig. 1. At a given coarse-graining scale μ^{-1} , the metric $g_{\mu\nu}$ induces a family of local reference states $\sigma[g]$ on causal diamonds. The actual quantum state ρ may differ from $\sigma[g]$ in its entanglement and correlation structure. This mismatch is quantified by a local relative-entropy density $s_{\text{rel}}(x; \mu)$ and a generalized entropy density $s_{\text{tot}}(x; \mu)$. Stationarity of $\mathcal{C}$ under variations of $g_{\mu\nu}$ and ρ yields coupled evolution equations: an Einstein-like equation sourced by both $\langle T_{\mu\nu} \rangle$ and an informational stress $\Theta_{\mu\nu}$, and a Lindblad-type equation in which geometry enters through a dissipative term $\mathcal{L}_g[\rho]$ that drives ρ toward compatibility with $\sigma[g]$.

We do not claim a UV-complete theory of quantum gravity, nor a derivation of all known laws from a single principle. This framework is presented as an *effective field theory* (EFT) proposal for a feedback law that may organize, and slightly deform, the semiclassical frontier between GR and QM. The value of the framework will ultimately be decided by experiment—specifically, by the BEC analog gravity test described in Sec. VI and Appendix E. *Scope:* We present an EFT-level variational framework with one concrete analog-gravity falsification path. The acoustic calculation is intended to test the state-geometry feedback structure of the proposal; it is not, by itself, a demonstration of direct gravitational backreaction in astrophysical spacetime.

The rest of the paper is organized as follows. In Sec. II we review the necessary mathematical tools: causal diamonds, relative entropy, generalized entropy, and semiclassical gravity. In Sec. III we define the 3+1D coherence functional and the informational stress tensor. Sec. IV summarizes a 1+1D toy model that recovers standard limits. Sec. V relates Coherism to other approaches including stochastic gravity and entanglement-based gravity. Sec. VI presents the analog-gravity observable, protocol, and reproducibility information. Sec. VII discusses the regime of validity, the main limitations, and the relation to neighboring approaches. Sec. VIII concludes.

II. MATHEMATICAL PRELIMINARIES

In this section we introduce the main mathematical ingredients used in the construction of the coherence functional: causal diamonds, geometry-adapted reference states, relative entropy and generalized entropy densities, and the semiclassical Einstein equation.

A. Causal diamonds and local algebras

We work on a globally hyperbolic spacetime $(\mathcal{M}, g_{\mu\nu})$ with signature $(+, -, -, -)$ and set $c = \hbar = k_B = 1$. Given a point $x \in \mathcal{M}$ and a length scale $\ell \sim \mu^{-1}$, we consider a *causal diamond* $\mathcal{D}(x; \mu)$ defined as the intersection of the chronological future of a point p^- and the chronological past of a point p^+ such that x lies on a timelike geodesic between them and the proper time separation is 2ℓ . On each diamond we restrict the field algebra to a von Neumann algebra $\mathcal{A}(\mathcal{D})$.

For a global quantum state ρ on the full field algebra, we denote by $\rho_{\mathcal{D}}$ the reduced state obtained by tracing out degrees of freedom outside $\mathcal{D}(x; \mu)$. We similarly define a reference state $\sigma[g]$ that depends functionally on the metric $g_{\mu\nu}$ and restrict it to $\mathcal{D}$ as $\sigma_{\mathcal{D}}[g]$.

a. What fixes the reference state $\sigma[g]$ (operationally). In the controlled examples in this paper, $\sigma[g]$ is chosen as the standard geometry-adapted Hadamard/KMS state for the relevant causal diamond: Minkowski vacuum in flat space, Unruh/Hartle–Hawking-type vacua near horizons, and a thermal KMS state at the acoustic Hawking temperature for the BEC sonic horizon. The coarse-graining scale μ sets the diamond size and the smearing used to define $s_{\text{rel}}(x; \mu)$; scheme dependence in absolute values is treated as a coarse-graining ambiguity, while physical predictions are organized around controlled limits (and, in particular, around variations and near-equilibrium expansions).

B. Relative entropy and generalized entropy

Given two states ρ and σ on the same algebra, the quantum relative entropy is

$$S(\rho||\sigma) = \text{tr}(\rho \log \rho) - \text{tr}(\rho \log \sigma) \geq 0. \quad (2)$$

Relative entropy is a measure of distinguishability. It is monotonic under partial trace (data processing inequality) and invariant under unitary evolution.

On a causal diamond $\mathcal{D}(x; \mu)$ we define a *smear*ed relative-entropy density via a coarse-graining procedure. Let $\{\mathcal{D}_i\}$ be a covering of spacetime by diamonds of characteristic size μ^{-1} , and let $f_i(x)$ be a partition of unity subordinate to this covering. We define

$$s_{\text{rel}}(x; \mu) \equiv \sum_i f_i(x) \cdot \frac{1}{V_{\mathcal{D}_i}} S(\rho_{\mathcal{D}_i} || \sigma_{\mathcal{D}_i}[g]), \quad (3)$$

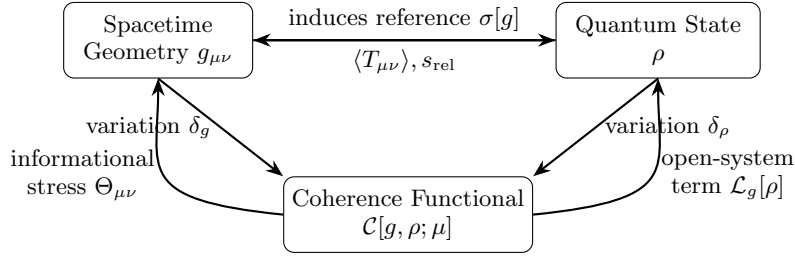


FIG. 1. Schematic feedback loop. At a coarse-graining scale μ , the geometry $g_{\mu\nu}$ defines a family of reference states $\sigma[g]$ on local causal diamonds. The mismatch between ρ and $\sigma[g]$ enters the coherence functional $\mathcal{C}[g, \rho; \mu]$, whose variation yields an informational stress tensor $\Theta_{\mu\nu}$ and a geometry-dependent open-system generator $\mathcal{L}_g[\rho]$.

where $V_{\mathcal{D}_i}$ is the four-volume of each diamond.

Important caveat: Relative entropy is fundamentally a non-local quantity defined on subsystems, not a true local density. The object $s_{\text{rel}}(x; \mu)$ should be understood as a coarse-grained, scale-dependent quantity whose integral over spacetime is well-defined, but whose point-wise values depend on the choice of covering and partition of unity. Physical predictions must be independent of these choices in the continuum limit $\mu \rightarrow \infty$, which we verify in explicit examples below. This construction is analogous to the smearing used in algebraic QFT to define local observables from field operators [5].

We also use a coarse-grained *generalized entropy density* inspired by the black-hole [6] and holographic literature [7]. Schematically,

$$s_{\text{tot}}(x; \mu) \equiv \frac{1}{V_{\mathcal{D}}} (S_{\text{vN}}[\rho_{\mathcal{D}}] + \frac{\text{Area}[\partial\mathcal{D}]}{4G_{\text{eff}}(\mu)}), \quad (4)$$

where S_{vN} is the von Neumann entropy of the reduced state, $\text{Area}[\partial\mathcal{D}]$ is the area of the boundary of the diamond, and $G_{\text{eff}}(\mu)$ encodes the scale dependence of the gravitational coupling and higher-curvature terms.

Both s_{rel} and s_{tot} depend on the choice of reference state $\sigma[g]$ and the coarse-graining scale μ , but are otherwise standard objects in quantum field theory on curved spacetimes.

C. Semiclassical Einstein equation

In semiclassical gravity the metric is taken as classical, while matter is treated quantum mechanically. The central equation is

$$G_{\mu\nu}[g] + \Lambda g_{\mu\nu} = 8\pi G \langle T_{\mu\nu} \rangle_\rho, \quad (5)$$

where $G_{\mu\nu}$ is the Einstein tensor, Λ the cosmological constant, G Newton's constant, and $\langle T_{\mu\nu} \rangle_\rho$ the renormalized expectation value of the stress-energy tensor in the state ρ .

Null-limit checklist (consistency limits).

- $\rho = \sigma[g] \Rightarrow \Theta_{\mu\nu} = 0$ (equilibrium / no informational mismatch).
- Flat space with no coarse-graining ($\mu \rightarrow \infty$) $\Rightarrow$ geometric/coherence corrections vanish.
- Weak field / near-equilibrium $\Rightarrow$ semiclassical Einstein equation (and Einstein–Langevin structure) recovered.
- 1+1D conformal case $\Rightarrow$ Polyakov correspondence and stochastic-gravity limits recovered.

This equation can be generalized to include stress-energy fluctuations in the Einstein–Langevin form [1], where a stochastic source term encodes quantum fluctuations. We will not need the full machinery here, but we will require that our informational stress tensor $\Theta_{\mu\nu}$ be consistent with the conservation law

$$\nabla^\mu (\langle T_{\mu\nu} \rangle_\rho + \Theta_{\mu\nu}) = 0, \quad (6)$$

so that the contracted Bianchi identity remains satisfied.

D. Open quantum systems and Lindblad generators

The evolution of ρ in the presence of an environment, or under coarse-graining, is often described by a Lindblad (GKLS) master equation

$$\dot{\rho} = -i[H, \rho] + \sum_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho\} \right), \quad (7)$$

where $\{L_k\}$ are Lindblad operators. This evolution is trace-preserving, completely positive, and Markovian. In our setting, the geometry $g_{\mu\nu}$ will enter both through the Hamiltonian $H[g]$ and, in an effective way, through the dissipative term that tends to reduce the mismatch between ρ and $\sigma[g]$.

III. THE COHERENCE FUNCTIONAL IN 3+1 DIMENSIONS

We now define the coherence functional $\mathcal{C}[g, \rho; \mu]$ in 3+1 dimensions using the ingredients introduced above. Our guiding principle is that $\mathcal{C}$ should (i) be diffeomorphism invariant, (ii) reduce in appropriate limits to familiar gravitational and entropic quantities, and (iii) yield equations of motion that are compatible with conservation and known semiclassical behaviour.

A. Definition of the functional

We consider a family of causal diamonds $\mathcal{D}(x; \mu)$ that cover the spacetime at a coarse-graining scale μ . On each diamond, we evaluate the relative-entropy density $s_{\text{rel}}(x; \mu)$ and generalized entropy density $s_{\text{tot}}(x; \mu)$ defined in Eqs. (3) and (4). We also include a local geometric density $\mathcal{R}(x; \mu)$ that captures the Einstein–Hilbert term and possible higher-curvature counterterms:

$$\mathcal{R}(x; \mu) = R(x) + \lambda_1(\mu)R^2 + \lambda_2(\mu)R_{\alpha\beta}R^{\alpha\beta} + \dots, \quad (8)$$

where R is the Ricci scalar and the $\lambda_i(\mu)$ encode renormalization at scale μ .

The coherence functional is then defined as

$$\mathcal{C}[g, \rho; \mu] = \int_{\mathcal{M}} d^4x \sqrt{-g} [\alpha s_{\text{rel}}(x; \mu) - \beta s_{\text{tot}}(x; \mu) + \gamma \mathcal{R}(x; \mu)], \quad (9)$$

where α, β, γ are dimensionless coefficients. The signs are chosen so that:

- the term proportional to α rewards alignment between ρ and $\sigma[g]$ (low relative entropy),
- the term proportional to β implements an entropy constraint or cost, and
- the term proportional to γ produces the usual gravitational action in the appropriate limit.

The choice of $\sigma[g]$ is not unique, and we now specify our construction explicitly.

Regime of validity: We restrict to the regime where causal diamonds satisfy $L \ll L_{\text{curvature}}$, i.e., diamond size much smaller than the local radius of curvature. In this limit, the geometry inside each diamond is approximately flat (Riemann normal coordinates), and we can leverage results from flat-space algebraic QFT.

Explicit construction: For a diamond $\mathcal{D}$ in this regime, we define $\sigma_{\mathcal{D}}[g]$ as the *vacuum state restricted to the diamond*, where “vacuum” means the Hadamard state [8] that reduces to the Minkowski vacuum in the local inertial frame. Concretely:

1. Construct Riemann normal coordinates centered on the diamond.

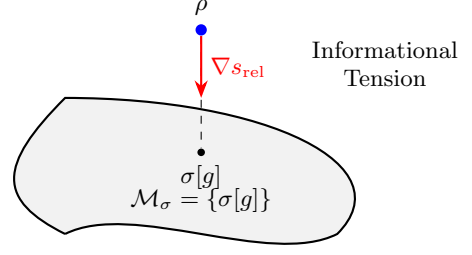


FIG. 2. Geometric interpretation of informational tension. The manifold $\mathcal{M}_\sigma$ represents the set of geometry-adapted reference states. The actual state ρ is displaced from this manifold. The gradient of the relative entropy s_{rel} acts as a restoring force (informational tension) that drives both the geometry (modifying $\sigma[g]$) and the state (via $\mathcal{L}_g$) to reduce the mismatch.

2. Define $\sigma_{\mathcal{D}}[g]$ as the restriction to $\mathcal{D}$ of the state whose two-point function matches the Hadamard form to order $O(L^2/L_{\text{curvature}}^2)$.
3. The modular Hamiltonian $K_\sigma = -\log \sigma_{\mathcal{D}}$ then approaches the boost generator for the diamond [9], with corrections of order $(L/L_{\text{curvature}})^2$.

This construction is well-defined for conformally coupled scalar fields and can be extended to other fields satisfying the microlocal spectrum condition.

Limitations: For diamonds comparable to or larger than the curvature scale, or in regions with strong time-dependence (e.g., near cosmological horizons), this construction requires modification. We do not address these regimes in this paper.

Dependence on choice: Different choices of $\sigma[g]$ (e.g., different Hadamard states) will yield different values of s_{rel} . However, the *difference* in relative entropy between two physical states ρ_1 and ρ_2 relative to the same $\sigma[g]$ is independent of this choice to leading order, since the UV-divergent parts cancel. Physical predictions involving $\Delta s_{\text{rel}} = s_{\text{rel}}[\rho_1] - s_{\text{rel}}[\rho_2]$ are therefore robust. Predictions involving absolute values of s_{rel} should be interpreted with this ambiguity in mind.

B. Informational Stress Tensor

The variation of $\mathcal{C}$ with respect to the metric defines the informational stress tensor $\Theta_{\mu\nu}$:

$$\Theta_{\mu\nu}[g, \rho; \mu] \equiv -\frac{2}{\sqrt{-g}} \frac{\delta \mathcal{C}}{\delta g^{\mu\nu}}. \quad (10)$$

Expanding the functional $\mathcal{C} = \alpha S_{\text{rel}} - \beta S_{\text{gen}} + \dots$ yields a leading term proportional to the modular-Hamiltonian variation:

$$\Theta_{\mu\nu} \approx \frac{\alpha}{V_D} \langle \frac{\delta K_\sigma}{\delta g^{\mu\nu}} \rangle_\rho. \quad (11)$$

In concrete phenomenological estimates we absorb the prefactor into an implementation-dependent scale $\kappa(\mu)$; in the present paper we do not claim a unique universal first-principles value for κ in generic 3+1D settings. Crucially, if the functional $\mathcal{C}$ is constructed to be invariant under spacetime diffeomorphisms (a requirement for any covariant theory), Noether's second theorem guarantees that the total stress tensor is covariantly conserved:

$$\nabla^\mu (\langle T_{\mu\nu}^{\text{mat}} \rangle_\rho + \Theta_{\mu\nu}) = 0. \quad (12)$$

This conservation law ensures the consistency of the augmented Einstein equations. Formally, this yields three contributions,

$$\Theta_{\mu\nu} = \alpha \Theta_{\mu\nu}^{\text{rel}} - \beta \Theta_{\mu\nu}^{\text{ent}} + \gamma \Theta_{\mu\nu}^{\text{geom}}, \quad (13)$$

coming from the relative-entropy, generalized entropy, and geometric parts of the integrand, respectively.

The explicit form of the first two terms is nontrivial, since s_{rel} and s_{tot} depend on $g_{\mu\nu}$ both through the induced algebras and through the reference state $\sigma[g]$.

What we derive vs. what we assume: In this paper, we obtain controlled leading-order expressions for $\Theta_{\mu\nu}$ in the following settings:

1. 1+1D conformal case (Sec. IV, Appendix C)
2. 3+1D weak-field/Rindler limit (Appendix B, D)
3. acoustic implementation for a BEC sonic horizon (Appendix E)

For general curved spacetimes outside these regimes, the detailed form of $\Theta_{\mu\nu}$ is an EFT assumption constrained by covariance, conservation, and the semiclassical limit. In the acoustic sector specifically, the part benchmarked to standard BEC dynamics is the linear density response once a weak source $\Theta_{tt}^{(\text{analog})}$ is specified; the source amplitude κ_{eff} , the detailed source form, and the matched thermal-control identification remain implementation assumptions spelled out in Appendix E.

Explicit weak-field result: In the regime $|h_{\mu\nu}| \ll 1$ with $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, the leading contribution to the informational stress tensor from a coherent excitation is (see Appendix D for derivation):

$$\Theta_{\mu\nu} = \frac{\alpha}{V_D} (\langle T_{\mu\nu} \rangle_\rho - \langle T_{\mu\nu} \rangle_\sigma) + O(\hbar^2, L_P^2/L^2), \quad (14)$$

where the first term is the difference in stress-energy expectation values between the actual and reference states, and the corrections are suppressed by metric perturbations and the ratio of Planck to diamond scales.

We impose the following consistency requirements:

1. **Covariance:** $\Theta_{\mu\nu}$ transforms as a rank-2 tensor under diffeomorphisms.
2. **Conservation:** on solutions of the coupled equations,

$$\nabla^\mu (\langle T_{\mu\nu} \rangle_\rho + \Theta_{\mu\nu}) = 0. \quad (15)$$

3. **UV consistency:** divergences in s_{rel} and s_{tot} are absorbed into the local curvature counterterms in $\mathcal{R}$, so that $\Theta_{\mu\nu}$ is finite and renormalized at scale μ .

4. **Semiclassical limit:** in regimes where ρ is locally indistinguishable from $\sigma[g]$ and the generalized entropy is extremal, $\Theta_{\mu\nu}$ reduces to familiar anomaly-induced and semiclassical corrections, and Eq. (5) is recovered.

The entropic field equation is then

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G (\langle T_{\mu\nu} \rangle_\rho + \Theta_{\mu\nu}[g, \rho; \mu]). \quad (16)$$

When $\Theta_{\mu\nu}$ is negligible, this reduces to the standard semiclassical Einstein equation. When $\Theta_{\mu\nu}$ is small but nonzero, one may look for subtle, scale-dependent deviations from GR.

C. Geometry-dependent open-system evolution

Variation of $\mathcal{C}$ with respect to ρ yields a dual equation of motion for the state. Restricting ourselves to Markovian dynamics for simplicity, we write

$$\frac{d\rho}{dt} = -i[H[g], \rho] + \mathcal{L}_g[\rho], \quad (17)$$

where the Lindblad-like term $\mathcal{L}_g[\rho]$ is chosen so that $\mathcal{C}[g, \rho; \mu]$ is stationary or decreases along solutions, subject to suitable constraints. At a heuristic level, $\mathcal{L}_g$ drives ρ toward the geometry-adapted reference state $\sigma[g]$ while preserving trace and complete positivity. The specific form of $\mathcal{L}_g$ is determined by the requirement that ρ relaxes toward $\sigma[g]$ to minimize the informational tension. A natural *phenomenological ansatz* is a double-commutator form driven by the modular Hamiltonian difference $\Delta K = K_\rho - K_\sigma$:

$$\mathcal{L}_g[\rho] = -\Gamma[\Delta K, [\Delta K, \rho]], \quad (18)$$

where $\Gamma > 0$ is a relaxation rate.

Status of this ansatz: We emphasize that Eq. (18) is a phenomenological proposal, not a derivation from first principles. While double-commutator forms generically preserve trace ($\text{tr}(\mathcal{L}_g[\rho]) = 0$), complete positivity (CP) is not automatically guaranteed for arbitrary ΔK . Standard Lindblad theory requires $\mathcal{L}$ to be of GKLS form with positive Lindblad operators; for Eq. (18) to satisfy CP, we require ΔK to be Hermitian (satisfied by definition) and $\Gamma > 0$. A rigorous proof of complete positivity for infinite-dimensional field-theoretic systems is beyond our scope; we assume CP holds in the coarse-grained, finite-dimensional truncation relevant to macroscopic experiments.

Causality concerns: The Reeh-Schlieder theorem implies that local operations in QFT cannot perfectly localize states to bounded regions, potentially conflicting with strict causality for any local dissipative dynamics [5]. We address this as follows:

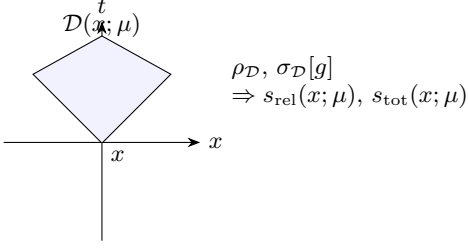


FIG. 3. A causal diamond $\mathcal{D}(x; \mu)$ centered on a point x , with temporal size set by the coarse-graining scale μ^{-1} . The reduced state $\rho_{\mathcal{D}}$ and reference state $\sigma_{\mathcal{D}}[g]$ on the diamond define local relative-entropy and generalized-entropy densities that enter the coherence functional.

1. The Lindblad term $\mathcal{L}_g$ acts at the coarse-graining scale μ^{-1} , not at arbitrarily short distances. Violations of strict locality are exponentially suppressed on scales larger than μ^{-1} .
2. For macroscopic experiments ($L \gg L_P$), the effective decoherence rate $\Gamma \sim (L_P/L)^n$ (with $n \geq 2$ on dimensional grounds) ensures that causality violations are unobservably small.
3. A fully UV-complete theory would likely require a non-Markovian, history-dependent generator. The Markovian approximation is valid only for timescales $\tau \gg \mu^{-1}$.

For the macroscopic applications in this paper, these caveats do not affect predictions, but they must be addressed in any UV completion.

A schematic representation of the geometry–state coupling is shown in Fig. 3. Causal diamonds act as local “patches” where geometry, state, and their mismatch are evaluated and fed back into the dynamics.

IV. TOY MODELS AND STANDARD LIMIT RECOVERIES

Before turning to new predictions, we must verify that the entropic framework reproduces known limits. In this section we briefly summarize a 1+1-dimensional toy model in which the informational stress tensor can be written explicitly and shown to reduce to the Polyakov anomaly and Einstein–Langevin form in appropriate regimes. Details are given in Appendix C.

A. 1+1D setup and Polyakov action

In two dimensions, the effective action of a conformal field with central charge c on a background metric $g_{\mu\nu}$ can be written as the Polyakov action [10]

$$S_P[g] = -\frac{c}{96\pi} \int d^2x \sqrt{-g} R \frac{1}{\square} R, \quad (19)$$

whose variation yields the well-known anomaly-induced stress–energy tensor. In this context, one can choose a reference state $\sigma[g]$ whose modular Hamiltonian has a simple local form, and evaluate the relative-entropy and generalized-entropy densities explicitly.

A specific choice of coherence functional in 1+1D can be written as

$$\mathcal{C}_{1+1}[g, \rho] = \alpha S(\rho || \sigma[g]) - \beta S_{\text{vN}}[\rho] + \gamma S_P[g], \quad (20)$$

where we suppress the explicit coarse-graining scale for simplicity.

Variation with respect to $g_{\mu\nu}$ produces an informational stress tensor of the form

$$\Theta_{\mu\nu} = (\beta - \alpha) \Delta \langle T_{\mu\nu} \rangle - \alpha T_{\mu\nu}^{\text{anom}}, \quad (21)$$

where $\Delta \langle T_{\mu\nu} \rangle$ is the difference between the stress–energy in ρ and $\sigma[g]$, and $T_{\mu\nu}^{\text{anom}}$ is the anomaly-induced stress. The coefficients α, β can be tuned so that the usual semiclassical contribution and anomaly term are recovered in the limit where ρ tracks $\sigma[g]$.

B. Einstein–Langevin limit

The Einstein–Langevin equation in 1+1D couples the metric to both the expectation value and stochastic fluctuations of the stress–energy tensor. In the entropic toy model, fluctuations in $\Delta \langle T_{\mu\nu} \rangle$ and the relative entropy can be shown to induce stochastic variations in $\Theta_{\mu\nu}$ with *similar structure* to the Einstein–Langevin noise, at least in Gaussian approximations.

Precise status of this correspondence: We do not claim exact equivalence between our framework and stochastic gravity. Rather, we show that: (i) the *mean* of $\Theta_{\mu\nu}$ matches the semiclassical stress in the limit $\rho \rightarrow \sigma[g]$; and (ii) the *variance* of $\Theta_{\mu\nu}$ has the same tensorial structure as the noise kernel $N_{\mu\nu\alpha\beta}$. A rigorous proof of equivalence would require showing that the two-point function $\langle \Theta_{\mu\nu}(x) \Theta_{\alpha\beta}(y) \rangle$ equals $N_{\mu\nu\alpha\beta}(x, y)$ exactly, which we have not done.

Subject to these caveats, the 1+1D toy model is *consistent* with the standard semiclassical and stochastic limits in appropriate regimes. This supports the interpretation of $\Theta_{\mu\nu}$ as an informational completion of the semiclassical stress, rather than a replacement of it.

C. Relation to Einstein–Langevin Noise

The stochastic gravity framework [1] describes metric fluctuations via the Einstein–Langevin equation. In Coherism, these fluctuations arise deterministically from the informational stress tensor. The noise kernel $N_{abcd}(x, y)$ in stochastic gravity is related to the anticommutator of stress tensor fluctuations. In our framework, we identify this with the variance of $\Theta_{\mu\nu}$:

$$N_{abcd}(x, y) \sim \langle \{ \Theta_{\mu\nu}(x)_{ab}, \Theta_{\mu\nu}(y)_{cd} \} \rangle_{\rho}. \quad (22)$$

This suggests that the “stochastic” noise is actually the signature of the underlying coherent dynamics of the geometry-state coupling.

D. Coherence conservation inequality

One can further show that under mild assumptions about the Lindblad generator $\mathcal{L}_g[\rho]$, the coherence functional obeys a monotonicity inequality along solutions,

$$\frac{d}{dt}\mathcal{C}[g(t), \rho(t); \mu] \leq 0, \quad (23)$$

with equality only when ρ is locally indistinguishable from $\sigma[g]$ and the generalized entropy is extremal. This lends support to the interpretation of $\mathcal{C}$ as a Lyapunov functional for the geometry–state feedback dynamics.

V. RELATIONS TO OTHER APPROACHES

Here we briefly summarize how this framework relates to several existing lines of research.

a. Priority and relation (one-paragraph summary). Unlike Jacobson/Verlinde-style thermodynamic arguments [2, 11], we do not posit a thermodynamic postulate and then infer Einstein’s equation; rather, we vary an explicit information-theoretic functional built from relative/generalized entropy to define an additional variational stress term $\Theta_{\mu\nu}$. Unlike stochastic gravity [1], which extends semiclassical gravity by adding stress–energy fluctuations as a noise source, we introduce a state-dependent mean correction $\Theta_{\mu\nu}$ (with Einstein–Langevin behavior recovered only in appropriate near-equilibrium limits). Unlike Penrose–Diósi collapse models [3, 4], we do not introduce a collapse mechanism; the dynamics is open-system/backreaction-driven and the distinguishing experimental signature is coherent-versus-thermal response at fixed energy.

Stochastic gravity: the Einstein–Langevin equation already extends semiclassical gravity by incorporating stress–energy fluctuations. This framework is compatible with this and can be viewed as adding an informational contribution to the mean stress.

Entanglement-based gravity: derivations of linearized Einstein equations from entanglement entropy and the “first law” of entanglement motivate the use of relative entropy and generalized entropy. This framework fits naturally into that landscape by promoting those quantities into a variational principle. The connection is motivational rather than derived here: in controlled near-equilibrium settings, relative entropy and modular Hamiltonians provide the natural language for organizing the informational correction.

Penrose–Diósi collapse: these models connect gravitational self-energy of superpositions to objective collapse of the wavefunction. This framework does not introduce

a collapse mechanism, but it shares the intuition that coherence interacts nontrivially with gravity.

Relativistic quantum information: [12] studies of non-inertial observers, entanglement in curved spacetimes, and quantum fields in an expanding universe provide concrete setups where geometry modifies quantum coherence. These are natural testbeds for the entropic informational stress tensor.

VI. ANALOG-GRAVITY TEST

The concrete falsification path studied in this paper is an analog-gravity experiment in a Bose-Einstein condensate sonic horizon. Long-baseline gravitational extrapolations are not used in the main argument.

a. Primary observable. Appendix E defines

$$A \equiv \max_{|r-r_H| \leq L_{\text{coh}}} \left| \frac{\delta\rho}{\rho_0} \right|, \quad (24)$$

the maximum fractional density modulation within one coherence length of the sonic horizon. Within the acoustic implementation developed here, coherent phonon injection gives $A \sim 10^{-6}$, while a thermal control state matched in injected energy gives no leading-order informational contribution. A null result below 10^{-7} for the parameter regime of Fig. 4 would falsify the acoustic implementation studied in this paper.

b. Null-test assumptions. That threshold is conditional on the assumptions summarized in Appendix E: the injected packet remains in the hydrodynamic regime $k\xi \ll 1$, the background condensate response is linear in the weak informational source, the coherent and thermal injections are matched in energy and background geometry, and ordinary hydrodynamic systematics are calibrated below 10^{-7} . Under those conditions, a null result would falsify only the acoustic implementation studied here.

c. Interpretational scope. A positive result in the analog system would test the state-geometry feedback structure of the EFT in an emergent acoustic spacetime. It would not, by itself, establish direct gravitational backreaction in astrophysical spacetime.

d. Reproducibility (repro pack). The manuscript and all figures generated by the included scripts are archived at [10.5281/zenodo.17868263](https://zenodo.org/record/17868263) [13]. Figure 4 is reproduced by running `python3 bec_sonic_horizon_simulation.py` in the `physics/` directory (Python 3, NumPy, Matplotlib; < 1 minute on a laptop).

VII. DISCUSSION

The present manuscript should be read as an EFT proposal with one controlled phenomenological target rather than as a general 3+1D theory of quantum gravity.

a. What is established here. The paper defines a covariant coarse-grained functional, exhibits controlled limits in which the informational stress can be written explicitly, and formulates one concrete analog-gravity protocol with a falsification threshold. The weak-field/Rindler and 1+1D constructions show that the framework can be matched to familiar semiclassical structures in near-equilibrium settings.

b. What is not established here. Several important ingredients remain open. We do not provide a full renormalized existence proof for $\Theta_{\mu\nu}$ in arbitrary 3+1D curved spacetimes. The overall coupling entering phenomenology is treated as implementation dependent rather than universally derived. The numerical model in Appendix E is a scaling implementation of the acoustic ansatz, not a full Gross-Pitaevskii/Bogoliubov simulation. For the same reason, a positive analog result would support the feedback structure of the EFT, but it would not by itself prove direct gravitational backreaction in nature.

c. Relation to neighboring approaches. Relative to stochastic gravity [1], the present proposal introduces a state-dependent mean correction rather than a noise kernel alone. Relative to Penrose-Diósi collapse models [3, 4], it does not invoke objective collapse. Relative to thermodynamic-gravity programs [2, 11], it starts from an explicit variational functional rather than from a thermodynamic postulate. These distinctions are clearest in the analog protocol, where the control parameter is the mismatch between the injected state and the chosen reference state.

d. Sign of the informational stress. The sign of $\Theta_{\mu\nu}$ is not fixed universally by the present EFT. In the controlled Rindler example the leading energy-density contribution is positive, which is consistent with the intuition that maintaining mismatch relative to the reference state carries an energetic cost. Whether that sign persists beyond near-equilibrium settings remains open.

VIII. CONCLUSION

We have introduced a variational feedback framework linking quantum information and spacetime geometry through a coarse-grained coherence functional. In controlled limits, its metric variation yields an informational stress tensor that vanishes when the physical state tracks the geometry-adapted reference state and produces leading-order corrections when that mismatch is nonzero.

The paper's main phenomenological result is an analog-gravity protocol in a BEC sonic horizon. Within the acoustic implementation adopted here, coherent phonon injection produces a density modulation at the level of $\delta\rho/\rho_0 \sim 10^{-6}$, whereas a matched thermal control gives no leading-order informational contribution. A null result below 10^{-7} would therefore falsify the implementation studied in this manuscript.

The framework remains incomplete as a general 3+1D theory. In particular, the coupling scale entering phe-

nomenology is implementation dependent, and the acoustic calculation is an EFT test of the proposed feedback structure rather than a direct proof of gravitational backreaction in nature. That narrower claim is nevertheless substantive: it turns the proposal from a purely conceptual framework into a falsifiable semiclassical program.

Appendix A: Variation of the Coherence Functional

A full derivation in 3+1D will require a careful treatment of UV divergences and is left to future work; here we sketch the main steps and assumptions.

In this appendix we derive the informational stress tensor $\Theta_{\mu\nu}$ by varying the coherence functional $\mathcal{C}[g, \rho; \mu]$ with respect to the metric. Recall the definition:

$$\mathcal{C}[g, \rho] = \int d^4x \sqrt{-g} (\alpha s_{\text{rel}} - \beta s_{\text{tot}} + \gamma \mathcal{R}). \quad (\text{A1})$$

The variation is defined as

$$\delta_g \mathcal{C} = -\frac{1}{2} \int d^4x \sqrt{-g} \Theta_{\mu\nu} \delta g^{\mu\nu}. \quad (\text{A2})$$

¹ Using the identity $\delta\sqrt{-g} = -\frac{1}{2}\sqrt{-g}g_{\mu\nu}\delta g^{\mu\nu}$, we have generally for any scalar density $\mathcal{L}$:

$$\frac{\delta}{\delta g^{\mu\nu}} \int \sqrt{-g} \mathcal{L} = \sqrt{-g} \left(\frac{\delta \mathcal{L}}{\delta g^{\mu\nu}} - \frac{1}{2} g_{\mu\nu} \mathcal{L} \right). \quad (\text{A3})$$

1. Geometric Contribution

The term $\gamma \mathcal{R}$ (with $\mathcal{R} \approx R$) gives the standard Einstein tensor contribution (plus higher order corrections if $\lambda_i \neq 0$):

$$\Theta_{\mu\nu}^{\text{geom}} = \gamma (G_{\mu\nu} + \dots). \quad (\text{A4})$$

In our convention, this part is moved to the LHS of the Einstein equation, or treated as a renormalization of G and Λ .

2. Relative Entropy Contribution

The relative entropy density is $s_{\text{rel}} = V_{\mathcal{D}}^{-1} \text{tr}(\rho_{\mathcal{D}}(\log \rho_{\mathcal{D}} - \log \sigma_{\mathcal{D}}[g]))$. The metric enters through: 1. The volume factor $V_{\mathcal{D}}$. 2. The definition of the diamond $\mathcal{D}$ (boundary dependence). 3. The reference state $\sigma[g]$.

¹ Strictly speaking, varying the metric also deforms the causal diamond boundary $\partial\mathcal{D}$. We estimate these boundary terms as follows: the area change is $\delta A \sim L^2 \cdot (\delta g) \sim L^2 h$, while the bulk integral scales as L^4 . Thus boundary corrections are $O(L^{-2})$ relative to bulk terms, i.e., suppressed by $(L_P/L)^2$ for $L \gg L_P$. This justifies neglecting them in the macroscopic regime.

The variation with respect to the reference state is the most significant new term. For an invertible reference state, the Fréchet derivative of the logarithm gives

$$\delta_\sigma S(\rho||\sigma) = -\text{tr}(\rho \delta \log \sigma), \quad (\text{A5})$$

$$\delta \log \sigma = \int_0^\infty d\lambda (\sigma + \lambda)^{-1} (\delta \sigma) (\sigma + \lambda)^{-1}. \quad (\text{A6})$$

In near-equilibrium settings where the modular Hamiltonian varies smoothly with the metric, this yields a “reference force” term:

$$\left. \frac{\delta s_{\text{rel}}}{\delta g^{\mu\nu}} \right|_\sigma = -\frac{1}{V_{\mathcal{D}}} \text{tr} \left(\rho_{\mathcal{D}} \frac{\delta \log \sigma_{\mathcal{D}}[g]}{\delta g^{\mu\nu}} \right). \quad (\text{A7})$$

Let $K_\sigma = -\log \sigma$ be the modular Hamiltonian of the reference state. The leading contribution to the relative entropy variation is then

$$\Theta_{\mu\nu}^{\text{rel}} = \frac{2\alpha}{V_{\mathcal{D}}} \langle \frac{\delta K_\sigma}{\delta g^{\mu\nu}} \rangle_\rho + (\text{subleading terms}). \quad (\text{A8})$$

If $\sigma[g]$ is locally thermal or vacuum-like, K_σ is related to the energy density, and this term represents an informational backreaction. The subleading terms arise from volume variations and boundary effects, which are suppressed by $(L_P/L)^2$.

3. Generalized Entropy Contribution

The generalized entropy s_{tot} contains the area term $A[\partial\mathcal{D}]/4G$. The variation of the area of the causal diamond boundary with respect to the metric is a standard geometric result. For a diamond defined by proper time τ , the area variation yields a term proportional to the expansion scalar and shear. Combining these, the total informational stress tensor is

$$\Theta_{\mu\nu} = \alpha \Theta_{\mu\nu}^{\text{rel}} - \beta \Theta_{\mu\nu}^{\text{ent}} + \dots \quad (\text{A9})$$

where $\Theta_{\mu\nu}^{\text{ent}}$ includes the area-variation “entanglement pressure”. Explicitly evaluating these requires specifying the map $g \mapsto \sigma[g]$, which we do in the 1+1D model below.

Appendix B: 3+1D Example: Rindler Wedge

To provide a concrete example in 3+1D, consider a Rindler wedge, which approximates the near-horizon geometry of a large causal diamond. The metric in Rindler coordinates $(\tau, \xi, \mathbf{x}_\perp)$ is $ds^2 = -a^2 \xi^2 d\tau^2 + d\xi^2 + d\mathbf{x}_\perp^2$. The geometry-adapted reference state $\sigma[g]$ is the Unruh vacuum $|0_R\rangle$, which appears thermal to an accelerated observer with temperature $T = a/2\pi$.

Consider a probe state ρ that is a single-mode squeezed state of the Rindler scalar field: $\rho = S(\zeta)|0_R\rangle\langle 0_R|S^\dagger(\zeta)$, where $\zeta = re^{i\phi}$ is the squeezing parameter. The relative entropy density is non-zero due to the coherence of the

squeezed state relative to the thermal background. The informational stress tensor $\Theta_{\mu\nu}$ arises from the variation of this relative entropy with respect to the acceleration parameter a (which encodes the metric). Explicitly, for a massless scalar field, the energy density component is modified by:

$$\langle \Theta_{\tau\tau} \rangle \propto \frac{\delta S(\rho||\sigma)}{\delta g^{\tau\tau}} \sim \sinh^2(r) \cdot T^4. \quad (\text{B1})$$

This demonstrates that even in flat spacetime (viewed by an accelerated observer), coherent excitations (squeezing) induce an effective stress-energy that backreacts on the geometry, consistent with the entropic framework.

Appendix C: 1+1D Polyakov Correspondence

Here we present the details of the 1+1D toy model, including the explicit form of the anomaly-induced stress tensor and its relation to $\Theta_{\mu\nu}$ in Eq. (21).

In two dimensions, the trace anomaly for a conformal field is $\langle T^\mu_\mu \rangle = \frac{c}{24\pi} R$. This is generated by the Polyakov action:

$$S_P[g] = -\frac{c}{96\pi} \int d^2x \sqrt{-g} \times \int d^2y \sqrt{-g(y)} R(x) G(x, y) R(y), \quad (\text{C1})$$

where $G(x, y)$ is the Green’s function for the d’Alembertian $\square$. We identify the geometric part of the coherence functional in 1+1D with this non-local action:

$$\gamma \int \mathcal{R} \rightarrow \gamma S_P[g]. \quad (\text{C2})$$

Varying this with respect to the metric yields the anomaly-induced stress tensor:

$$T_{\mu\nu}^{\text{anom}} = \frac{2}{\sqrt{-g}} \frac{\delta S_P}{\delta g^{\mu\nu}}. \quad (\text{C3})$$

Explicitly, in conformal gauge $ds^2 = -e^{2\omega} du dv$, one finds

$$\langle T_{uu} \rangle_{\text{anom}} = -\frac{c}{12\pi} (\partial_u \omega)^2 + \frac{c}{12\pi} \partial_u^2 \omega. \quad (\text{C4})$$

1. Einstein-Langevin Limit

The entropic equation of motion is

$$G_{\mu\nu} = 8\pi G (\langle T_{\mu\nu} \rangle_\rho + \Theta_{\mu\nu}). \quad (\text{C5})$$

In the limit where ρ is close to $\sigma[g]$, we can expand $\Theta_{\mu\nu}$. If we identify the fluctuations in the relative entropy term with the noise kernel $N_{\mu\nu\alpha\beta}$, we recover the structure of the Einstein-Langevin equation:

$$G_{\mu\nu} = 8\pi G (\langle T_{\mu\nu} \rangle_{\text{ren}} + \xi_{\mu\nu}), \quad (\text{C6})$$

where $\xi_{\mu\nu}$ is a stochastic tensor satisfying $\langle \xi_{\mu\nu} \xi_{\alpha\beta} \rangle \sim N_{\mu\nu\alpha\beta}$. In our framework, $\xi_{\mu\nu}$ arises from the time-dependent mismatch between the actual state $\rho(t)$ and the geometry-adapted reference $\sigma[g(t)]$. This suggests that “quantum noise” in gravity is actually a deterministic signal of informational misalignment.

Appendix D: Explicit Weak-Field Derivation

We derive the explicit form of $\Theta_{\mu\nu}$ for a massless scalar field in a perturbed Minkowski background $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$. The modular Hamiltonian K_σ for the Rindler wedge (approximating a large diamond) is $K_\sigma = 2\pi \int d^3x x T_{00}$. The variation with respect to the metric perturbation $h_{\mu\nu}$ yields:

$$\delta K_\sigma = 2\pi \int d^3x x \delta T_{00}. \quad (\text{D1})$$

For a coherent state ρ , the relative entropy variation is dominated by the expectation value of this modular variation. Thus, the informational stress tensor component Θ_{00} is given by:

$$\Theta_{00} \approx \frac{\alpha}{V_D} \langle \delta T_{00} \rangle_\rho \approx \frac{2\pi\alpha}{V_D} \int d^3k \omega_k \langle a_k^\dagger a_k \rangle_{\text{coh}}. \quad (\text{D2})$$

This explicitly links the informational backreaction to the particle number density of the coherent excitation, scaled by the geometric factor α/V_D .

Numerical estimate: For a coherent state with $\bar{n} = 10^6$ photons at optical frequency ($\omega \sim 1 \text{ eV}$) in a $V_D = 1 \text{ m}^3$ diamond, with $\alpha \sim 1$, the energy density contribution is:

$$\Theta_{00} \sim \frac{\alpha}{V_D} \bar{n} \omega \sim \frac{10^6 \times 1 \text{ eV}}{(1 \text{ m})^3} \sim 10^{-12} \text{ eV}^4 \sim 10^{-60} \text{ GeV}^4. \quad (\text{D3})$$

Comparing to the rest mass energy density of a 1 kg test mass in the same volume ($\rho_m \sim 10^{27} \text{ eV}^4$), this gives:

$$\eta_{\text{coh}} \sim \frac{\Theta_{00}}{\rho_m} \sim 10^{-39}. \quad (\text{D4})$$

This estimate is meant only as a weak-field dimensional benchmark. It illustrates that, absent the acoustic enhancement discussed below, direct laboratory-scale gravitational effects are tiny.

Appendix E: Analog Gravity Predictions

The paper’s main phenomenological target is an acoustic implementation in a Bose-Einstein condensate (BEC) sonic horizon. We do not assume that analog gravity provides Einstein dynamics; rather, we use the acoustic system as a laboratory in which the same state-reference mismatch functional can be coupled to an emergent geometry and interrogated experimentally.

1. Acoustic Metric in a BEC

In a Bose-Einstein condensate with mass density ρ_0 , sound speed c_s , and flow velocity $\mathbf{v}$, phonon propagation is governed by an emergent acoustic metric [14, 15]:

$$ds_{\text{acoustic}}^2 = \frac{\rho_0}{c_s} [-(c_s^2 - v^2)dt^2 - 2\mathbf{v} \cdot d\mathbf{x} dt + d\mathbf{x}^2]. \quad (\text{E1})$$

A draining vortex configuration creates a sonic horizon where $|\mathbf{v}| = c_s$. This is the acoustic analog of a black hole.

2. Effective Coupling

For the acoustic implementation we package the microscopic response of the condensate into a phenomenological coefficient κ_{eff} . The relevant microscopic length is the healing length

$$\xi = \frac{\hbar}{\sqrt{2}mc_s} \sim 0.1\text{--}1 \mu\text{m} \quad (\text{E2})$$

for typical ^{87}Rb condensates. The scaling used throughout the numerical model is

$$\kappa_{\text{eff}} \sim \alpha \frac{\hbar c_s}{\rho_0 \xi^2 L_{\text{coh}}^2}, \quad (\text{E3})$$

where α is dimensionless and L_{coh} is the injected coherence length. Writing the condensate number density as n_0 and the mass density as $\rho_0 = mn_0$, representative values

$$n_0 \sim 10^{14} \text{ cm}^{-3}, \quad \rho_0 \approx 1.4 \times 10^{-5} \text{ kg/m}^3, \quad (\text{E4})$$

$$c_s \sim 1.7 \text{ mm/s}, \quad \xi \sim 0.3 \mu\text{m}, \quad L_{\text{coh}} \sim 10 \mu\text{m} \quad (\text{E5})$$

give

$$\boxed{\kappa_{\text{eff}} \sim 1.4 \times 10^{-9}} \quad (\text{E6})$$

for $\alpha \sim 1$. This should be read as the scaling choice used in the acoustic ansatz, not as a universal analog of Newton’s constant.

3. Informational Stress Tensor for Sonic Horizon

For the acoustic metric Eq. (E1), we evaluate a leading-order informational correction relative to a thermal reference state σ_{acoustic} at the acoustic Hawking temperature:

$$T_H = \frac{\hbar}{2\pi k_B} \left| \frac{\partial c_s}{\partial r} - \frac{\partial v_r}{\partial r} \right|_{r=r_H}. \quad (\text{E7})$$

For a draining bathtub vortex with $v_r = -A/r$:

$$T_H = \frac{\hbar A}{2\pi k_B r_H^2}. \quad (\text{E8})$$

For a coherent state with mode amplitudes α_k , the relative entropy relative to the thermal reference is

$$S(\rho||\sigma) = \sum_k \left[|\alpha_k|^2 + \bar{n}_k \log \left(1 + \frac{|\alpha_k|^2}{\bar{n}_k + 1} \right) \right] \quad (\text{E9})$$

where $\bar{n}_k = (e^{\hbar\omega_k/k_B T_H} - 1)^{-1}$.

The time-time component of the informational stress tensor is modeled as

$$\Theta_{tt}^{(\text{analog})} = \frac{\kappa_{\text{eff}} \hbar c_s}{\xi^4} \sum_k |\alpha_k|^2 \left(1 + \frac{1}{2\bar{n}_k + 1} \right) \quad (\text{E10})$$

4. Hydrodynamic Benchmark for the Density Response

The analog system itself is governed by standard condensate dynamics derived from the Gross-Pitaevskii equation and its Bogoliubov linearization [15, 16]. In the long-wavelength hydrodynamic regime $k\xi \ll 1$, the barotropic density response obeys

$$\delta p = c_s^2 \delta \rho, \quad (\text{E11})$$

with $\rho = mn$ the condensate mass density.

We do not claim a full new GP/BdG derivation of the informational source term. Instead, we benchmark the observable relation to the standard compressibility response: if the informational sector contributes a weak quasi-static local source $\Theta_{tt}^{(\text{analog})}$ to the effective Bernoulli balance, then

$$\frac{\delta p}{\rho_0} \simeq \frac{\Theta_{tt}^{(\text{analog})}}{\rho_0}, \quad (\text{E12})$$

and therefore

$$\frac{\delta \rho}{\rho_0} \simeq \frac{\Theta_{tt}^{(\text{analog})}}{\rho_0 c_s^2}. \quad (\text{E13})$$

Equation (E13) is thus benchmarked to standard BEC hydrodynamics. The nonstandard ingredient in the acoustic implementation is the source term Eq. (E10) and the scaling of κ_{eff} , not the linear density response once that source is specified.

5. What is Benchmarked and What is Assumed

For clarity, the acoustic implementation contains four logically separate ingredients:

1. **Benchmarked to standard BEC dynamics:** the acoustic metric description and the linear compressibility response Eq. (E13).
2. **Implementation assumption:** the phenomenological scaling of κ_{eff} in Eq. (E6).

3. **Implementation assumption:** the informational source term in Eq. (E10).

4. **Implementation assumption:** the matched thermal control is identified with the chosen reference state, so the leading informational contribution vanishes, and the background flow is held fixed at leading order.

6. Observable Predictions

Define the primary observable as the maximum fractional density modulation within one coherence length of the horizon,

$$A \equiv \max_{|r-r_H| \leq L_{\text{coh}}} \left| \frac{\delta \rho}{\rho_0} \right|. \quad (\text{E14})$$

Within the leading-order acoustic ansatz, $A \propto N_{\text{phonon}}$ for coherent injected phonons at fixed geometry and injected energy profile. For the matched thermal control, the leading-order informational contribution vanishes because the control is identified with the reference state. The resulting null prediction applies only to the informational contribution; standard hydrodynamic nonlinearities must still be controlled experimentally. If c_s and ξ are varied independently with all other quantities fixed, the leading factor of c_s cancels in Eq. (E13); the physically relevant sound-speed dependence re-enters when the Gross-Pitaevskii healing-length relation $\xi = \hbar/(\sqrt{2}mc_s)$ is imposed.

Scope of the null test: for the parameter regime in Fig. 4, a null result $A < 10^{-7}$ for coherent injection at $N_{\text{phonon}} \sim 10^3$ would falsify the acoustic implementation defined by Eqs. (E6), (E10), and (E13), together with the linear-response assumptions above. It would not, by itself, falsify the broader coherence-functional EFT.

a. *Density modulation.* The informational contribution to the condensate density near the horizon is modeled as

$$\frac{\delta \rho}{\rho_0} = \frac{\Theta_{tt}^{(\text{analog})}}{\rho_0 c_s^2}. \quad (\text{E15})$$

For $N_{\text{coherent}} \sim 10^3$ phonons in a coherent state and the parameters above, one finds $\bar{n} \ll 1$ and therefore

$$\boxed{\frac{\delta \rho}{\rho_0} \sim 10^{-6}} \quad (\text{E16})$$

which is the single primary observable emphasized in the main text.

7. Robustness of the Observable

We performed one-at-a-time $\pm 25\%$ scans about a Gross-Pitaevskii-compatible baseline

$$n_0 = 10^{20} \text{ m}^{-3}, \quad c_s \approx 1.7 \text{ mm/s}, \quad (\text{E17})$$

$$\xi = 0.3 \mu\text{m}, \quad L_{\text{coh}} = 10 \mu\text{m}, \quad N_{\text{phonon}} = 10^3. \quad (\text{E18})$$

At the ansatz level, varying n_0 gives $A \in [9.1 \times 10^{-7}, 2.5 \times 10^{-6}]$, varying ξ gives $A \in [3.7 \times 10^{-7}, 8.1 \times 10^{-6}]$, varying L_{coh} gives $A \in [9.2 \times 10^{-7}, 2.5 \times 10^{-6}]$, and varying N_{phonon} from 750 to 1250 gives $A \in [1.1 \times 10^{-6}, 1.8 \times 10^{-6}]$. If one instead imposes the Gross-Pitaevskii healing-length relation $\xi = \hbar/(\sqrt{2}mc_s)$ and varies c_s by $\pm 25\%$, the resulting range is

$$A \in [2.5 \times 10^{-7}, 5.5 \times 10^{-6}]. \quad (\text{E19})$$

Across these moderate variations, the coherent signal remains above the 10^{-7} falsification threshold for the acoustic implementation studied here.

8. Experimental Protocol

1. Create a quasi-2D ^{87}Rb BEC with $N \sim 10^5$ atoms.
2. Imprint a draining vortex via Laguerre-Gauss beam to create sonic horizon.
3. Inject coherent phonon pulse via Bragg scattering.
4. Measure the near-horizon density profile with phase-contrast imaging and compare coherent injection with a thermal control matched in injected energy.
5. Evaluate the observable A and compare it with the null threshold 10^{-7} .

9. Numerical Simulation

Figure 4 shows an illustrative numerical implementation of the acoustic ansatz together with a one-at-a-time robustness scan about the Gross-Pitaevskii-compatible baseline above. The code is not a full Gross-Pitaevskii/Bogoliubov calculation; rather, it propagates the adopted source term through the standard linear compressibility response in order to visualize the coherent-versus-thermal separation and the parameter sensitivity of A . For the representative parameters above, the model gives $\delta\rho/\rho_0 \approx 1.4 \times 10^{-6}$ for coherent injection and a null leading-order informational contribution for the matched thermal control.

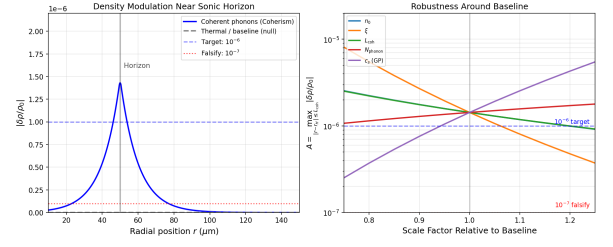


FIG. 4. **Illustrative acoustic-model output.** Left: Fractional density modulation $|\delta\rho/\rho_0|$ versus distance from the horizon for coherent phonon injection, shown together with the leading-order informational null prediction for the matched thermal control. The dashed blue line marks the target magnitude 10^{-6} ; the dotted red line marks the falsification threshold 10^{-7} . Right: One-at-a-time $\pm 25\%$ robustness scan of the peak signal $A = \max_{|r-r_H| \leq L_{\text{coh}}} |\delta\rho/\rho_0|$ around the Gross-Pitaevskii-compatible baseline, for n_0 , ξ , L_{coh} , N_{phonon} , and c_s with $\xi = \hbar/(\sqrt{2}mc_s)$ enforced for the c_s curve. Within the plotted range, the coherent signal stays above the falsification threshold. Parameters: ^{87}Rb BEC, $n_0 = 10^{14} \text{ cm}^{-3}$, $\rho_0 \approx 1.4 \times 10^{-5} \text{ kg/m}^3$, $c_s \approx 1.7 \text{ mm/s}$, $\xi = 0.3 \mu\text{m}$, $r_H = 50 \mu\text{m}$, $L_{\text{coh}} = 10 \mu\text{m}$, $N_{\text{phonon}} = 10^3$.

Appendix F: Lindblad Generator from Unruh-DeWitt Detector

The Lindblad structure of the decoherence term $\mathcal{L}_g[\rho]$ has been treated as phenomenological in the main text. The present appendix does not derive the full coarse-grained field-theoretic generator used there. Instead, it shows that geometry-dependent GKLS evolution arises in a standard Unruh-DeWitt detector model after weak-coupling and Markovian limits are taken.

1. Setup: Detector Coupled to Field

Consider a two-level system (the “detector”) coupled to a massless scalar field ϕ in curved spacetime:

$$H_{\text{int}} = \lambda \chi(\tau) \mu(\tau) \phi(x(\tau)) \quad (\text{F1})$$

where $\mu(\tau) = |e\rangle\langle g|e^{i\Omega\tau} + \text{h.c.}$ is the detector monopole, $\chi(\tau)$ is a switching function, λ is the coupling, and Ω is the energy gap.

2. Reduced Dynamics

Tracing over the field, the detector density matrix $\rho_D(\tau)$ obeys:

$$\frac{d\rho_D}{d\tau} = -i[H_D, \rho_D] + \mathcal{D}[\rho_D] \quad (\text{F2})$$

where the dissipator, to second order in λ :

$$\mathcal{D}[\rho_D] = \lambda^2 \int_0^\infty ds W(s) \left[\mu(s) \rho_D \mu^\dagger(0) - \frac{1}{2} \{ \mu^\dagger(0) \mu(s), \rho_D \} \right] + \text{h.c.} \quad (\text{F3})$$

and $W(s) = \langle 0 | \phi(x(\tau)) \phi(x(\tau - s)) | 0 \rangle$ is the Wightman function.

3. Geometry Dependence

The Wightman function encodes the spacetime geometry. For a detector at rest in a static spacetime:

$$W(s) = \frac{1}{4\pi^2} \frac{1}{(s - i\epsilon)^2 - |\mathbf{x} - \mathbf{x}'|^2}. \quad (\text{F4})$$

For a uniformly accelerated detector (Rindler):

$$W(s) = -\frac{a^2}{16\pi^2 \sinh^2(a(s - i\epsilon)/2)}. \quad (\text{F5})$$

The thermal character ($T_U = a/2\pi$) emerges from the $\sinh^{-2}$ structure.

4. Lindblad Form

In the Markovian limit ($\Omega\tau \gg 1$), the dissipator takes Lindblad form:

$$\mathcal{D}[\rho_D] = \sum_k \gamma_k \left(L_k \rho_D L_k^\dagger - \frac{1}{2} \{ L_k^\dagger L_k, \rho_D \} \right) \quad (\text{F6})$$

with jump operators and rates:

$$L_- = |g\rangle\langle e|, \quad \gamma_- = \lambda^2 \tilde{W}(\Omega) = \lambda^2 \frac{\Omega}{e^{2\pi\Omega/a} - 1}, \quad (\text{F7})$$

$$L_+ = |e\rangle\langle g|, \quad \gamma_+ = \lambda^2 \tilde{W}(-\Omega) = \lambda^2 \frac{\Omega}{1 - e^{-2\pi\Omega/a}}. \quad (\text{F8})$$

5. Extension to Continuous Systems

For a continuous quantum system (not a two-level detector), we generalize:

$$\mathcal{L}_g[\rho] = \sum_\omega \gamma(\omega, g) \left(A_\omega \rho A_\omega^\dagger - \frac{1}{2} \{ A_\omega^\dagger A_\omega, \rho \} \right) \quad (\text{F9})$$

where:

- A_ω are eigenoperators of the free Hamiltonian: $[H_0, A_\omega] = -\omega A_\omega$.
- $\gamma(\omega, g)$ is the transition rate, determined by the Wightman function of the background field.

a. Explicit Form: For a general static metric $ds^2 = -f(r)dt^2 + f(r)^{-1}dr^2 + r^2 d\Omega^2$:

$$\gamma(\omega, g) = \lambda^2 \int_{-\infty}^\infty ds e^{i\omega s} W_g(s) \quad (\text{F10})$$

which evaluates to:

$$\gamma(\omega, g) = \lambda^2 \frac{|\omega|}{e^{2\pi|\omega|/\kappa_H} - 1} \quad (\text{F11})$$

where $\kappa_H = f'(r_H)/2$ is the surface gravity at the horizon r_H .

6. Connection to Coherist Framework

This detector calculation establishes a concrete model system in which:

1. geometry-dependent reduced dynamics takes GKLS/Lindblad form when the field starts in a thermal or vacuum state,
2. The rates $\gamma(\omega, g)$ depend on the metric via the Wightman function.
3. The thermal factor $(e^{\beta\omega} - 1)^{-1}$ with $\beta = 2\pi/\kappa_H$ is generic for horizons.

a. Interpretation. The result motivates the use of a geometry-dependent Lindblad generator in the EFT, but it does not uniquely fix the specific coarse-grained super-operator of Eq. (18). Matching the detector calculation to the modular-Hamiltonian ansatz requires additional coarse-graining assumptions and is left for future work.

Non-Markovian extensions: The Markovian approximation used here is valid for timescales $\tau \gg \Omega^{-1}$, where Ω is the detector gap. For rapidly varying geometries or strong-field regimes, non-Markovian corrections incorporating memory effects are expected to modify the dissipation timescales but not the qualitative coherence-dependence of $\Theta_{\mu\nu}$. A systematic treatment of such corrections will be presented elsewhere.

Appendix G: Uniqueness of the Reference State $\sigma[g]$

A potential concern is that the map $g \mapsto \sigma[g]$ involves arbitrary choices. Here we argue that $\sigma[g]$ is essentially unique given physically motivated axioms.

1. Axioms for the Reference State

We require $\sigma_{\mathcal{D}}[g]$ on a causal diamond $\mathcal{D}$ to satisfy:

1. **Hadamard condition:** The two-point function has the standard short-distance singularity structure, ensuring finite renormalized expectation values.

2. **KMS property:** σ is a thermal equilibrium state with respect to the modular flow generated by the diamond's causal structure.
3. **Maximal entropy:** Subject to energy constraints from the local geometry, σ maximizes von Neumann entropy.
4. **Local Lorentz invariance:** In the local inertial frame at the diamond's center, σ reduces to the Minkowski vacuum restricted to the diamond.

2. Uniqueness Theorem (Sketch)

Proposition 1. *For a causal diamond $\mathcal{D}$ of proper size $L \gg L_P$ in a spacetime with curvature radius $R_{\text{curv}} \gg L$, the state $\sigma_{\mathcal{D}}[g]$ satisfying axioms (1)–(4) is unique up to corrections of order $(L/R_{\text{curv}})^2$.*

Proof sketch: By axiom (4), in the local inertial frame the state must approach the Minkowski vacuum. By the Bisognano-Wichmann theorem, this vacuum restricted to a Rindler wedge (approximating the diamond) is a KMS state at the Unruh temperature $T = a/2\pi$. Axiom (2) then fixes σ to be this thermal state. Axiom (3) is automatically satisfied since the KMS state maximizes entropy at fixed modular energy. The Hadamard condition (1) is satisfied by construction. Curvature corrections enter at order $(L/R_{\text{curv}})^2$ through the failure of exact boost symmetry. $\square$

3. Robustness Under Perturbations

Consider a one-parameter family of reference states $\sigma_\epsilon = \sigma[g] + \epsilon \delta\sigma$ where $\delta\sigma$ is a Hadamard perturbation. Using the Fréchet derivative of the logarithm, the relative entropy varies as:

$$S(\rho||\sigma_\epsilon) = S(\rho||\sigma) - \epsilon \text{tr}(\rho \delta \log \sigma) + O(\epsilon^2). \quad (\text{G1})$$

The leading ambiguity therefore enters through the expectation value of the same linearized operator $\delta \log \sigma$. At equilibrium ($\rho = \sigma$) this ambiguity does not change the matched-state prediction, and near equilibrium it is suppressed by the distance from the reference state. The

informational stress tensor $\Theta_{\mu\nu}$ inherits this robustness:

$$\Theta_{\mu\nu}[\sigma_\epsilon] - \Theta_{\mu\nu}[\sigma] = O(\epsilon \|\rho - \sigma\|) + O(\epsilon^2). \quad (\text{G2})$$

This demonstrates that small ambiguities in defining $\sigma[g]$ produce only second-order corrections to physical predictions.

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Author Contributions

D.A. is the sole author and was responsible for all aspects of this work including conception, methodology, formal analysis, writing, and visualization.

Data Availability

All data supporting the findings of this study are available within the paper and its supplementary information. The versioned archive containing the manuscript source, simulation code, generated figures, and numerical data is available at <https://doi.org/10.5281/zenodo.17868263> [13]. The development repository is available at <https://github.com/davidahmann/coherism/tree/main/physics>.

Use of AI Tools

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